

Mohammad Hajjo

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SUMMARY

Computer Science Engineer specialized in embedded systems, IoT, and real-time applications. Experienced in building end-to-end systems, including an ESP32-based handwriting recognition solution and distributed fault-tolerant networks. Strong in C/C++ and Python with hands-on experience in hardware diagnostics, low-level debugging, and system optimization. Focused on building efficient, reliable systems that perform under real-world constraints.

TECHNICAL SKILLS

Programming Languages: Java, Python, NetLogo, JavaScript, C, C++, C#, Rust Assembly, MATLAB

Embedded Systems Frameworks: ESP-IDF, FreeRTOS, STM32Cube HAL, Zephyr RTOS, Mbed OS, Micropython

Libraries & Tools: Git, Docker, Visual Studio Code, Linux, SQL, Node.js, Bootstrap, Figma

Concepts: Multithreading, Concurrency, Real-Time Systems, Wireshark, Agile Development, Scrum

EDUCATION

Malmö University

B.S. of Science in Engineering in Computer Science

Aug 2022 – Jun 2025

Malmö, Sweden

Thesis: [Thesis published on DiVA Portal with 1000+ downloads](#)

EXPERIENCE

Production/Machine Operator

Aimpoint

Feb 2026 – Present

Malmö, Sweden

- Performed quality assurance testing on optical systems, ensuring compliance with strict performance and safety standards.
- Assembled precision optical and mechanical components with high accuracy following technical specifications.
- Interpreted technical drawings and SOPs to maintain production consistency and reduce assembly errors.

Internship

Nobel Teknik

Sep 2025 – Feb 2026

Malmö, Sweden

- Repaired and maintained smartphones, tablets, and computers, including both iPhone and Android devices.
- Diagnosed and resolved hardware and software issues efficiently, ensuring high customer satisfaction.
- Performed component-level repairs and soldering to restore hardware functionality without full part replacements.

Teaching Assistant

Malmö University

Aug 2023 – Jun 2024

Malmö, Sweden

- Assisted in embedded systems and programming labs, guiding students through coding, debugging, and hardware integration.
- Guided students in electronics, microcontrollers, and system design through individual support and feedback.
- Troubleshoot complex hardware/software issues during practical sessions, ensuring smooth lab progress.

Private Tutor

Studentea (org nr: 969792-3028)

Nov 2023 – Jun 2024

Malmö, Sweden

- Delivered personalized tutoring in mathematics, physics, and programming, adapting methods to individual learning styles.

- Supported students with test preparation, assignments, and conceptual understanding, leading to improved performance.
- Maintained strong student relationships, resulting in extended contracts and positive feedback.

Supervisor

Eleda Stadion

Jun 2021 – May 2025

Malmö, Sweden

- Oversaw kiosk operations during large-scale events, ensuring smooth service for thousands of visitors.
- Managed staff scheduling, task delegation, and on-site problem solving to maintain efficiency and customer satisfaction.
- Coordinated inventory logistics and enforced compliance with food safety and hygiene standards across multiple units.

PROJECTS

Letter Recognition for Handwriting on Embedded System Using a Machine Learning Model

Thesis Project

Spring 2025

Python, PyTorch, ESP-IDF, ML, Electronics

- Built a real-time handwriting recognition system on ESP32-CAM using a custom-written CNN.
- Optimized model with quantization and data augmentation to meet embedded memory and performance constraints.
- Developed a fully standalone, low-power embedded system with on-device image capture, processing, and display.

Multi-Threaded Chat System

Software Development Project

Spring 2024

Java, Networking, Threads, OOP

- Developed a client-server chat application supporting multiple concurrent users via socket programming.
- Implemented message broadcasting and connection management with robust error handling.
- Applied object-oriented design principles to structure communication protocols and thread synchronization.

Firefighter Coordination System

Software Development Project

Fall 2024

ESP32, C, Networking, Threads, Embedded Systems

- Designed and implemented a peer-to-peer mesh communication system on ESP32 devices using ESP-NOW.
- Enabled real-time status updates, task assignment, and location sharing across multiple nodes without central infrastructure.
- Optimized communication reliability and latency for large-scale deployments in emergency scenarios.

Multi-Threaded Chat System

Distributed Intelligent System

Fall 2024

Distributed Systems, ML, Networking, NetLogo, Git

- Designed and implemented a multi-node system with agents exchanging information over message queues / gRPC, coordinating tasks in a distributed environment.
- Incorporated machine-learning modules for decision making / prediction at edge-nodes with limited compute, balancing load and accuracy.
- Handled fault tolerance: node failures, retries, message ordering, consistency, etc.

HOBBY PROJECTS

Automated IC Identification & Validation System

Automated system project

Winter 2025

C/C++, Arduino, SPI, UI/UX, Hardware Debugging, QA

- Engineered a high-reliability diagnostic system using Arduino Mega and a ZIF-socket to automate digital IC identification.
- Developed a custom Search Algorithm based on pin signatures, significantly reducing lookup time vs. sequential methods.
- Programmed high-speed SPI communication for SD card database interfacing and real-time TFT LCD processing.
- Implemented "Validate Mode" for pin-level diagnostics, detecting gate failures often missed by commercial tools.

- Designed an interactive GUI with a virtual keyboard to enhance user experience and visual feedback.

IoT-Enabled Smart Plant Watering System

Embedded Systems Project

Fall 2025
ESP32, IoT, Sensors, C/C++,
Blynk

- Designed and built an autonomous plant watering system using ESP32 with capacitive soil moisture, temperature, and humidity sensors.
- Controlled a water pump via MOSFET driver based on real-time soil and environmental conditions to prevent overwatering.
- Integrated IoT functionality for remote monitoring and manual control using the Blynk platform.
- Implemented data logging, adaptive watering logic, and power-efficient deep sleep mode for outdoor operation.

Smart Restaurant Management and Ordering System

Full-Stack Mobile Application Project

Fall 2025
Flutter, SQLite, UI/UX, PDF
Generation

- Developed an interactive restaurant configuration app allowing users to draw the restaurant layout and assign numbered tables.
- Implemented a menu management system to add, edit, and price menu items dynamically.
- Designed a cashier interface enabling waiters to select tables, take orders, and print detailed receipts.
- Utilized SQLite for persistent storage of layouts, menus, and order history with modular architecture for scalability.

ADDITIONAL INFORMATION

Driving licence: B (Manual)

Forklift licence: Certified operator

Languages: Swedish (native), English (fluent), Arabic (native), Turkish (intermediate)

Certifications:

- C# for .NET Developers
- Agile Development and Scrum
- Google Tools of the Trade: Linux and SQL
- Google Foundations: Data, Data, Everywhere
- Microsoft Secure Access with Azure Active Directory
- Software Engineering: Modeling Software Systems using UML